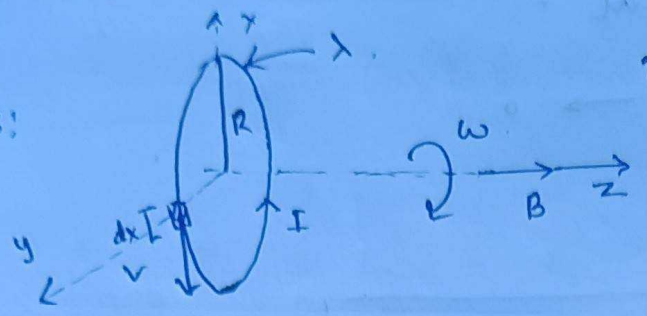
Tutorial - 2

(Q) A charged ring of radius R with charge density λ is rotating about its axis with ω. Calculate $\bar{B}$ @ the center of the ring.

Ans:



$$I_{ring} = \frac{dQ}{dt}$$

$$= \cancel{\frac{d}{dt}(\lambda 2\pi}$$

$$T = \frac{2\pi}{\omega}$$

The charge complete 1 full rotation in T.

$$I = \frac{Q}{T} = \frac{\lambda(2\pi R)}{2\pi/\omega} = \lambda\omega R$$

Alternative:

Consider a small element dx of ring.

$$dq = \lambda\, dx$$

$$\Rightarrow \frac{dq}{dt} = \lambda \frac{dx}{dt} = \lambda v = \lambda R\omega = I$$

From Ampere's Law:

$$\oint B\, d\bar{l} = \mu_0 I$$

$$\Rightarrow B \oint dl = \mu_0(\lambda R\omega)$$

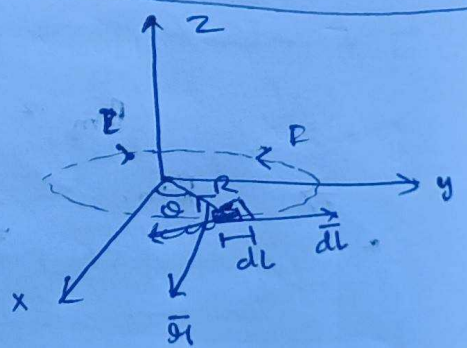
$$\Rightarrow B(2\pi R) = \mu_0(\lambda R\omega)$$

[$\bar{B} \parallel d\bar{l}$ on ring & constant @ fixed distance R]

Biot - Savart's Law:

$$\bar{B} = \frac{\mu_0}{4\pi} \int R \frac{d\bar{l} \times \hat{r}}{r^2}$$

$$\bar{r} = R\cos\theta\, i + R\sin\theta\, j$$



$\therefore \hat{r} = \cos\theta\, i + \sin\theta\, j$

$d\bar{l} \times \hat{r} = dl \cancel{\sin\theta} \sin 90^\circ = dl\, \hat{k}$ $[\bar{r} \perp d\bar{l}]$

$\hat{k}$

$\therefore \bar{B} = \frac{\mu_0}{4\pi} \int I \times \frac{dl}{r^2} \hat{k}$

$= \frac{\mu_0}{4\pi R^2} I \oint_0^{2\pi R} dl = \frac{\mu_0 I}{4\pi R^2} \cdot 2\pi R = \frac{\mu_0 I}{2R} \hat{k}$

$\therefore \bar{B}_{center} = \frac{\mu_0 I}{2R} \hat{k}$

Substituting the values of I:

$\bar{B}_c = \frac{\mu_0 (\lambda R \omega)}{2R} \hat{k} = \boxed{\frac{\mu_0 \lambda \omega}{2} \hat{k} = \bar{B}_c}$ (Ans)